

Mohith Nagendra

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Education

University of Maryland, College Park – BS in Computer Science - Machine Learning

May 2025

Minor: Entrepreneurial Leadership

GPA: 3.72

QUEST Honors Program - Consulting for **Meta**, Dean's List, BigTh!nk AI, and Google Developer Student Club

MIT Pre-LGO Pipeline Program

Skills

Languages: Java, Kotlin, Python, Go, SQL, TypeScript, JavaScript, C++, C#, Rust, Scala, Shell Scripting, Swift

Streaming & Distributed Systems: Apache Kafka, Amazon Kinesis, Amazon SQS, event-driven architecture, delivery semantics (retries/DLQ/idempotency), distributed consistency, on-call incident response

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, CUDA, NumPy, Pandas, Apache Spark, Ray, RAPIDS, OpenCV

Infra & Systems: AWS, GCP, Docker, Kubernetes, Terraform, CloudFormation, CDK, Redis, PostgreSQL, Datadog, CI/CD, Git

Open Source: Arctic Code Vault Contributor; enhanced *nba-api* with new endpoint (3,000+ stars)

Experience

Software Development Engineer, Amazon

August 2025 – Present

- Unlocked **seven-figure annual GMS** impact through new priority-based pricing strategies, by cellularizing our core JVM-based pricing engine service – the central brain of the pricing system – into a fault-isolated, cell-based architecture
- Cut campaign planning time by 84% (3 days to 4 hours per campaign), saving **250+ hours** (1.5 months) per country manager annually across 12+ campaigns globally, through a multi-agent orchestration system with human-in-the-loop review gates that integrates with existing pricing levers and product offerings.
- Processed **120K+ price overrides** to date, enabling business teams to competitively price and run pricing experiments in real time, by building a serverless event-driven system on Lambda, DynamoDB, SQS, and SNS
- Sustained system stability during high-volume launch events and redrives, by implementing autoscaling and rate limiting to manage backpressure, while powering real-time repricing via a custom CDC system that reacts to upstream changes

Software Engineer I, Allergan Data Labs (AbbVie)

September 2024 – May 2025

- Drove a **73% higher conversion rate** over prior generic campaigns, by building the data pipeline to power a self-service tool for clinic managers to launch personalized campaigns, with idempotent delivery pulling patient account data from internal data lakes
- Increased user **offer redemption by 1.26M+** through embeddings-based ML recommendation models that personalize offers, with drift monitoring and a feedback pipeline for iterative retraining
- Eliminated unreviewed infrastructure changes across the data pipeline, by codifying provisioning in Terraform and Helm charts to enforce code review on every change

Machine Learning Engineer Intern, Oracle

June 2024 – September 2024

- Improved fraud detection accuracy by **27%** on MySQL-backed transaction graphs, through a Graph Neural Network detection system with bias audits across demographic segments to ensure equitable model behavior
- Cut ticket resolution time from 2–3 days to **1 day** across enterprise-scale ticket volume, by deploying a Graph RAG resolution system over internal support docs using Pinecone and GPT-4
- Achieved **97%** accuracy on a flight ETA prediction model, through multi-source feature engineering and model ensembling

Software Engineer Intern, Allergan Data Labs (AbbVie)

May 2024 – August 2024

- Increased user retention by 31% through owning the design-to-launch of the *Rewards Reminder* campaign, reaching **6.5M+ users**
- Enabled proactive issue detection across the pipeline, from data gathering to deliverability, by building Datadog monitors & alarms

Full Stack Engineer, NitesOut

November 2023 – May 2025

- Scaled a React Native ticketing app to **30K+ users**, eliminating service fees and cutting prices by 27%, through leading a team of 4 across the platform's build and launch
- Drove attendance up through Stripe API integration (**\$50K+ monthly transactions**) and automated Firebase messaging

Machine Learning Engineer Intern, Airweb Digital

November 2023 – May 2024

- Extracted vehicle flow and density metrics across 100+ hours of drone footage and 12+ cities, by implementing a traffic analysis pipeline with OpenCV and PyTorch
- Reduced manual ad-placement workflow time by **90%** through developing an automated spatial scoring and visibility ranking model with map-based artifacts

Full Stack Engineer, Cornell Tech (Break Through Tech)

December 2023 – March 2024

- Achieved 95% positive user feedback with **sub-200ms** response times (500+ RPM), through shipping *SussanahAI*, an NLP web app using the Cohere API to analyze tone, summarize content, and filter spam via a Flask backend and React frontend

Genome Computing Project Manager and Lead Researcher, UMD FIRE

August 2022 – August 2023

- Improved DNA-drug interaction analysis accuracy by **25%** through leading a 7-member research team modeling in Python and R; reduced genomic data processing execution time by **65%** through optimizing workflows in C++ with multithreaded parallel computing

Software Engineering and Robotics Mentor, FIRST

September 2023 – Present